DocLens: Executive Summary & Hackathon Judges Guide

Comprehensive Evaluation Briefing & Technical Overview for Devpost Judges

Document Standard: Devpost Hackathon Submission Artifact | Author: DocLens Engineering Team

1. Executive Overview & Vision Statement

DocLens is an enterprise-ready, AI-assisted forensic signature verification application designed to deliver calibrated assessments, multi-reference specimen comparison, pixel-level visual evidence, and explainable forensic narrative reports. In high-stakes legal, financial, banking, and insurance transactions, signatures authorize critical decisions including contracts, wire transfers, property deeds, and corporate compliance forms.

When a signature is disputed or questioned, reviewers currently face a difficult dilemma: either rely on manual visual examination by forensic document examiners, which is slow, expensive, and subject to personal bias, or rely on traditional machine learning models that output black-box match scores (e.g. "87% authentic") with zero inspectable visual evidence. DocLens addresses this problem by treating signature verification not as an isolated black-box prediction, but as a completely explainable forensic analysis workflow.

The core philosophy driving DocLens is straightforward: when an AI-assisted assessment matters in a critical workflow, the reviewer must be able to follow the empirical visual evidence behind it. DocLens therefore unifies document ingestion, automatic signature localization, deep neural feature extraction, classical stroke dynamics measurement, writer-disjoint score calibration, Grad-CAM visual difference heatmaps, and LLM-assisted narrative generation into one seamless application.

2. The Forensic Problem Statement & Market Opportunity

2.1 The Global Signature Verification Challenge

Signature forgery remains one of the primary vectors for financial fraud, unauthorized contract execution, and identity impersonation. Millions of legal contracts and banking documents are processed daily across financial institutions. When a signature authenticity dispute arises, document reviewers must compare the questioned signature against multiple genuine reference specimens while accounting for natural biometric handwriting variation, stroke thickness changes, baseline drift, and pen-lift patterns.

In legal proceedings, audit reviews, and corporate compliance checks, simply presenting a numerical match percentage is insufficient. Reviewers need inspectable visual proof highlighting where stroke discrepancies occur, whether stroke width variations align with genuine specimens, and whether pen-lift gaps indicate deliberate tracing or genuine fluid writing execution.

3. Comprehensive Comparison: Traditional vs DocLens Approach

Evaluation Metric / Capability	Manual Human Forensic Review	Traditional Black-Box AI	DocLens Forensic Platform
Verification Execution Speed	24 to 72 Hours per case	< 5 Seconds automated	< 3.5 Seconds automated
Inspectable Visual Evidence	Manual photographic notes	None (Opaque score only)	Pixel-level Grad-CAM & Stroke Markers
Multi-Reference Consensus	Subjective visual comparison	Single reference matching	Multi-reference centroid consensus

Document Ingestion	Manual scanning & cropping	Manual cropped images only	Multi-page PDF & Image Auto-Crop
Score Calibration	Qualitative opinion	Uncalibrated probabilities	Frozen CEDAR Writer-Disjoint Model
Audit Report Generation	Manual typing (hours)	None / Simple score text	Instant Exportable Vector PDF Report

4. Core Technical Architecture & Innovation

DocLens combines deep neural representation learning with classical computer-vision stroke measurements to ensure maximum scoring reliability and forensic transparency.

4.1 Deep Neural Visual Representation (PyTorch ResNet18)

DocLens passes normalized 224x224 RGB signature crops through a PyTorch ResNet18 visual feature extractor. The final classification layer is removed, yielding a high-dimensional 512-dimensional embedding vector. L2 normalization is applied to convert visual features into a normalized spatial representation where cosine distance directly measures visual stroke resemblance.

4.2 Classical Computer-Vision Structural Dynamics

To complement deep neural representations, DocLens calculates five quantitative structural metrics using OpenCV image analysis: Stroke Width Variation (distance transform distributions), Letter Proportion Inconsistency (connected component aspect ratios), Baseline Deviation (horizontal writing axis estimation), Pen Lift Positions (stroke gap spatial coordinates), and Slant Angle Deviation (principal axis rotational orientation).

4.3 Writer-Disjoint Calibrated Decision Engine

DocLens applies a frozen ensemble decision calibration derived from the benchmark CEDAR signature dataset under a strict writer-disjoint evaluation protocol. The final decision score combines 95% Deep Embedding Similarity and 5% Classical Structural Similarity evaluated against a calibrated decision threshold of 0.9271469.

5. Technical Architecture Pipeline Flow

The complete DocLens execution pipeline moves through 14 distinct automated stages:

- **Stage 1 - Case Creation:** User creates an authenticated case workspace specifying Case Title and Signer Name.
- **Stage 2 - Material Upload:** User uploads 1 questioned document (image or multi-page PDF) and 1 to 5 genuine reference specimens.
- **Stage 3 - PDF Candidate Selection:** PyMuPDF scans multi-page PDFs at 200 DPI, scores text density, and extracts optimal signature candidate representations.
- **Stage 4 - Signature Auto-Localization:** OpenCV bounding-box detector locates candidate signature regions without manual user cropping.
- **Stage 5 - Image Normalization:** Signature crops are normalized to RGB, EXIF oriented, constrained to safety limits, and resized to 224x224.
- **Stage 6 - PyTorch Feature Extraction:** ResNet18 trunk operating in PyTorch inference_mode() extracts 512D L2-normalized embeddings.
- **Stage 7 - Multi-Reference Comparison:** Cosine similarity is computed against all reference specimens and the genuine centroid profile.

- **Stage 8 - Structural Feature Measurements:** OpenCV measures stroke width variance, letter proportions, baseline drift, pen lifts, and slant angle.
- **Stage 9 - Calibrated Score Calculation:** Ensemble score (95% deep / 5% structural) is evaluated against frozen 0.9271469 CEDAR threshold.
- **Stage 10 - Grad-CAM Activation Mapping:** Gradient-weighted class activation maps compute pixel-level visual difference heatmaps.
- **Stage 11 - LLM Narrative Synthesis:** Structured findings are sent to Groq Llama-3.3-70b to generate a plain-English explanation.
- **Stage 12 - Deterministic Fallback Engine:** If Groq API is offline, system automatically generates a structured template report.
- **Stage 13 - Web Dashboard Rendering:** React UI renders interactive HTML5 Canvas evidence view and structural metric breakdown cards.
- **Stage 14 - Vector PDF Export:** jsPDF client library generates a standalone, court-ready PDF case report with QR verification.

6. Empirical Benchmark Performance & Statistical Validation

DocLens enforces rigorous scientific evaluation standards. The signature verification pipeline was benchmarked using the standard CEDAR dataset with 55 total writers, 1,320 genuine samples, and 1,320 skilled forgery samples under a writer-disjoint split (44 calibration writers, 11 unseen test writers).

Benchmark Metric	Calibration Writer Split (44 Writers)	Unseen Test Writer Split (11 Writers)
Writer Disjoint Split	44 Writers (80% dataset)	11 Writers (20% held-out test)
Pairwise Comparisons	1,760 Pairwise Evaluated Pairs	440 Unseen Test Pairs
Balanced Accuracy	93.11%	96.50% (High generalization)
ROC Area Under Curve (AUC)	Baseline Standard	99.59% (Near-perfect discrimination)
Equal Error Rate (EER)	Baseline Standard	3.41% EER
False Acceptance Rate (FAR)	5.40%	3.03% (Forgeries accepted)
False Rejection Rate (FRR)	8.38%	3.98% (Genuine rejected)

7. Hackathon Judging Criteria Alignment Matrix

- **Technical Difficulty & Execution:** Full-stack integration of FastAPI asynchronous Python server, PyTorch ResNet18 visual embeddings, OpenCV contour analysis, PyMuPDF candidate selection, Grad-CAM activation mapping, Groq Llama-3.3 LLM narrative engine, and Supabase Auth/DB/Storage.
- **Innovation & Explainability:** Pioneers explainable forensic signature AI by rendering pixel-level Grad-CAM visual heatmaps, stroke markers, and plain-English narrative reports grounded strictly in empirical findings.

- **UI/UX Design Excellence:** React 18 SPA featuring modern dark mode, HTML5 Canvas evidence overlays, interactive structural metric breakdown cards, multi-case management, and instant vector PDF report export.
- **Real-World Commercial Viability:** Directly applicable to banking compliance, legal document auditing, contract verification, and insurance fraud detection with zero setup overhead.

8. Step-by-Step Live Demonstration Guide for Judges

Judges evaluating DocLens live at https://doclens.yvesseraphin.xyz can execute an end-to-end audit following these steps:

- **Step 1 - Access & Authentication:** Navigate to doclens.yvesseraphin.xyz and log in or register a new user account. Authentication is secured via Supabase JWT bearer tokens.
- **Step 2 - Case Workspace Creation:** Click "New Case", enter a Case Title (e.g. "Banking Audit #4092") and Signer Name. This creates a multi-tenant case record in PostgreSQL.
- **Step 3 - Material Ingestion:** Upload a questioned document (JPEG, PNG, WebP, or multi-page PDF up to 40 pages) and 1 to 5 genuine reference specimen signatures.
- **Step 4 - Automated Signature Analysis:** Click "Analyze Signature". The backend automatically localizes the signature bounding box, extracts 512D ResNet18 embeddings, computes 5 classical structural features, and calculates calibrated scores against the 0.9271469 CEDAR threshold.
- **Step 5 - Evidence Inspection & PDF Export:** Review the "Visual Evidence" tab to inspect Grad-CAM heatmaps, baseline indicators, and pen-lift markers. Click "Export PDF" to generate a standalone court-ready forensic report.

9. Frequently Asked Questions (FAQ) for Technical Auditors

- **Q1: Does DocLens replace human forensic document examiners?** No. DocLens is designed as an AI-assisted decision support system. It provides calibrated metrics, visual evidence, and structured summaries to accelerate human review and ensure objective documentation.
- **Q2: How does DocLens handle multi-page PDF contracts?** DocLens includes a dedicated PyMuPDF candidate selection stage that renders PDF pages at 200 DPI, inspects text density, scores embedded images, and automatically selects the optimal page for signature localization.
- **Q3: What happens if the Groq LLM API is unavailable?** DocLens includes a deterministic template fallback engine. If the LLM API times out or fails, the system automatically generates a structured plain-text narrative preserving all computed numerical metrics and verdicts.
- **Q4: How is multi-tenant security enforced?** Multi-tenant security is enforced at the database level using Supabase PostgreSQL Row-Level Security (RLS) policies checking auth.uid() = user_id on all queries.

10. Hackathon Artifact Directory & Submission Package

The complete DocLens hackathon submission package includes the following artifacts:

- **Live Production Web Application:** https://doclens.yvesseraphin.xyz
- **GitHub Source Code Repository:** https://github.com/yvesseraphin/DocLens
- **Executive Judges Guide (This PDF):** docs/DocLens_Executive_Judges_Guide.pdf
- **Software Requirements Specification (SRS):** docs/DocLens_Software_Requirements_Specification.pdf
- **CEDAR Benchmark & ML Evaluation Report:** docs/DocLens_CEDAR_Benchmark_Report.pdf
- **Developer API Specification:** docs/DocLens_API_Specification.pdf
- **48-Hour Development Process Log:** docs/DocLens_48Hour_Development_Log.pdf